INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS553)

LAB 8

Assignment 8: Classes and Objects

Assignment aim: Define classes, fields and methods, then create objects.

Experiment No. 8.1 — Create and display one Student

1. Experiment Title: Create and display one Student

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A class defines fields and methods; an object is an instance with its own field values.

5. Algorithm / Procedure:
Define Student fields and show(); create an object; assign field values; call show().
Then compile and execute the Java program.

6. Program:

```
class Student {
    int roll;
    String name;
    void show() {
        System.out.println("Roll: " + roll);
        System.out.println("Name: " + name);
    }
}
public class Task_01_Create_and_display_one_Student {
    public static void main(String[] args) {
        Student s = new Student();
        s.roll = 5;
        s.name = "Rohan";
        s.show();
    }
}
```

7. Input: Fixed input values are declared in the program: Student roll/name or Book page values specified in the task.

8. Output:
Roll: 5
Name: Rohan

Time / Space Complexity: O(1) time; O(1) auxiliary space for this fixed example.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is a class?
- What is an object?
- What is a method?

Experiment No. 8.2 — Create and display two Students

1. Experiment Title: Create and display two Students

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A class defines fields and methods; an object is an instance with its own field values.

5. Algorithm / Procedure:
Create two Student objects; set each object's fields; call show() on both objects.
Then compile and execute the Java program.

6. Program:

```
class Student {
    int roll;
    String name;
    void show() {
        System.out.println("Roll: " + roll);
        System.out.println("Name: " + name);
    }
}
public class Task_02_Create_and_display_two_Students {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.roll = 5; s1.name = "Rohan";
        Student s2 = new Student();
        s2.roll = 8; s2.name = "Meera";
        s1.show();
        s2.show();
    }
}
```

7. Input: Fixed input values are declared in the program: Student roll/name or Book page values specified in the task.

8. Output:
Roll: 5
Name: Rohan
Roll: 8
Name: Meera

Time / Space Complexity: O(1) time for two fixed objects; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- Can two objects have different field values?
- What does new do?
- How is a method called on an object?

Experiment No. 8.3 — Check whether a Book is long

1. Experiment Title: Check whether a Book is long

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: A class defines fields and methods; an object is an instance with its own field values.

5. Algorithm / Procedure:
Define pages and isLong(); create two books with different page counts; call isLong() and print each boolean result.
Then compile and execute the Java program.

6. Program:

```
class Book {
    int pages;
    boolean isLong() {
        return pages > 300;
    }
}
public class Task_03_Check_whether_a_Book_is_long{
    public static void main(String[] args) {
        Book b1 = new Book(); b1.pages = 450;
        Book b2 = new Book(); b2.pages = 120;
        System.out.println(b1.isLong());
        System.out.println(b2.isLong());
    }
}
```

7. Input: Fixed input values are declared in the program: Student roll/name or Book page values specified in the task.

8. Output:
true
false

Time / Space Complexity: O(1) time; O(1) auxiliary space for two fixed objects.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is a boolean?
- What does return do?
- Why is 450 considered long?